Name _____ Period _____ Date _____

The Puppy Puzzle

Focus question: How can puppies from the same parents be alike, but not identical?

Notice and wonder

Compare the traits

Glue in: Puppy litter data

Name _____ Period _____ Date _____

The Puppy Puzzle (continued)

My first model

Draw: Parent 1, Parent 2, and two offspring. Use arrows to show information moving from the parents to the offspring. Label one shared trait and one different trait.

Three possible explanations

Three small sketch boxes: one idea in each.

Key idea

Exit ticket

Name _____ Period _____ Date _____

Mixing Information

Focus question: If offspring get information from the same two parents, why might each get a different combination?

Warm-up

Make possible offspring

Glue in: Card table: 3 offspring

Name _____ Period _____ Date _____

Mixing Information (continued)

Analyze the pattern

Revise our model

Revise your Day 1 model.

Three sources of variation

Name _____ Period _____ Date _____

Genetics Notes, Part 1: Genes and Alleles

Focus question: What are genes and alleles, and what do dominant and recessive mean?

Warm-up

Heredity

Chromosomes

Genes make proteins make traits

Name _____ Period _____ Date _____

Genetics Notes, Part 1: Genes and Alleles (continued)

Alleles

Sketch a pair of chromosomes. Color each set of alleles a different color.

Dominant vs. recessive

Black or brown cats

Draw three cat heads: BB, Bb, bb. Color each one to show its coat.

Exit ticket

Name _____ Period _____ Date _____

Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares

Focus question: How can a Punnett square show the possible allele combinations?

Warm-up

Genotype and phenotype

Draw two eyes. Color one blue and one brown.

Glue in: Genotype to phenotype: pea plants

Name _____ Period _____ Date _____

Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares (continued)

Homozygous and heterozygous

Write CC, Ee, Gg, dd, AA, Hh. Color the homozygous ones green and the heterozygous ones orange. Then write the missing words from the slide.

Punnett squares

Glue in: Punnett squares: steps and worked example

Practice

Exit ticket

Name _____ Period _____ Date _____

Monster Genetics, Part 1: Flip a Coin, Build a Monster

Focus question: How can two coin flips build a monster's genotype?

Warm-up

How the lab works

Glue in: Monster Genetics: the coin rule and Table 2

Name _____ Period _____ Date _____

Monster Genetics, Part 1: Flip a Coin, Build a Monster (continued)

Build the female monster

Use Table 1 on your Monster Genetics sheet.

Fill in the male monster

Use Table 2 (glued in above) and your Monster Genetics sheet.

Compare

Exit ticket

Name _____ Period _____ Date _____

Monster Genetics, Part 2: Predict the Baby

Focus question: How do sex cells and fertilization mix alleles, and how can we predict the baby?

Warm-up

Sex cells and fertilization

Punnett squares for the baby

Glue in: Punnett square: skin color and eyes

Name _____ Period _____ Date _____

Monster Genetics, Part 2: Predict the Baby (continued)

Percent chance

Class data

Glue in: Prediction vs. class count

Name _____ Period _____ Date _____

When Copying Goes Wrong

Focus question: How do copying errors and UV light change genes, and when are the changes inherited?

Warm-up

How do mutations happen?

Four kinds of mutations

Name _____ Period _____ Date _____

When Copying Goes Wrong (continued)

UV and mutations

Glue in: UV light and mutations

Who passes it on?

Glue in: Who passes it on? Five cases

Explain it

Name _____ Period _____ Date _____

Where Did This Variation Come From?

Focus question: How can data show where a genetic variation came from?

Clues for each source

Case 1: the puppy litter

Glue in: Case data: three cases

Name _____ Period _____ Date _____

Where Did This Variation Come From? (continued)

Case 2: a new base

Use “Case data: three cases” (glued in above).

Case 3: X-rays and mice

Use “Case data: three cases” (glued in above).

Write your claim

Name _____ Period _____ Date _____

Back to the Puppies and Make-Up Work

Focus question: How do I defend my claim against another explanation?

How to defend a claim

The new coat pattern

Glue in: The new coat pattern: two kennels

Name _____ Period _____ Date _____

Back to the Puppies and Make-Up Work (continued)

Fix my Day 1 model

Redraw your Day 1 model. Add: new combinations, copying errors, mutations from the environment, and which changes are inherited.

Write It: the puppy puzzle

Make up missing work

Name _____ Period _____ Date _____

Review Day: Prepare for the CSA

Focus question: Which CFA reviews will best get me ready for the CSA?

Check my CFAs

Glue in: My target tracker

My reviews

Name _____ Period _____ Date _____

Review Day: Prepare for the CSA (continued)

Enrichment: environment and traits

Glue in: Hydrangea soil data

Self-check

Rate each target 1 to 4 on your tracker (glued in above).